

CSCE 642 DEEP REINFORCEMENT LEARNING · MODULE 02 · LECTURE NOTES

Exploration

You cannot learn about what you never try.

MODULE OUTCOMES

Explain the bandit problem and why it is the right starting point.

Compare the standard exploration strategies.

Explain optimism in the face of uncertainty.

Explain why deep exploration is harder than action noise.

Explain intrinsic motivation and its failure mode.

1 Bandits

A BANDIT IS REINFORCEMENT LEARNING WITH THE SEQUENTIAL PART REMOVED

k actions, each with an unknown reward distribution, no state, and no consequences carried forward. Pull an arm, observe a reward, repeat — and nothing you did earlier changes what the actions now do.

So the *only* difficulty is exploration. Which makes the bandit the right place to study exploration in isolation, and it is the one part of this subject with substantial theory and provably near-optimal algorithms.

The measure is *regret*: the cumulative difference between the reward you obtained and the reward the single best fixed action would have given over the same horizon. **Logarithmic regret is the best achievable,** and §1's table says which strategies attain it.

And a great deal of real applied work is a bandit rather than a full MDP — which advert to show, which layout to serve, which of five treatments to assign. **If your actions have no lasting consequences for the environment's state, do not use an MDP** (Module 01 §4): a contextual bandit is far easier, has theory, and will work.

Strategy	How it works	Property
ϵ -greedy	Act greedily, but take a uniformly random action with probability ϵ .	Linear regret at a fixed ϵ — because it keeps exploring forever at a constant rate. Decay ϵ, and the Robbins–Monro reasoning of CSCE 669 Module 08 §2 applies to the schedule.
Optimistic initialisation	Initialise every action's estimated value well above any achievable reward.	Free, requires no extra machinery, and is surprisingly effective — see §2's explanation of why.

Strategy	How it works	Property
Upper confidence bound (UCB)	Choose the action maximising mean + $c\sqrt{(\ln N / n)}$, where n is that action's count and N the total.	Logarithmic regret — provably near-optimal. The same formula as CSCE 625 Module 05 §3's MCTS selection step , because it is the same problem.
Thompson sampling	Maintain a posterior over each action's value, sample one value from each, and act greedily on the samples.	Logarithmic regret, and frequently the best in practice — and it is the cleanest expression of §2's principle.
Boltzmann / softmax	Sample actions with probability proportional to $\exp(Q/\tau)$.	The temperature τ is hard to set , because the right value depends on the scale of Q , which changes during learning.

2 Optimism

OPTIMISM IN THE FACE OF UNCERTAINTY

Act as though every uncertain option is as good as it plausibly could be. Then one of two things happens: the option really is good, and you gain reward; or it is not, and you learn that, which is information you needed.

Either way the optimism is self-correcting — an overestimate survives only until the action is tried enough times — **and the agent never persistently avoids something it has not tried. Which is exactly ϵ -greedy's real failure:** it explores uniformly at random, so it keeps retrying actions it has already established are bad, and gives a never-tried action no more attention than a known-bad one.

UCB implements the principle explicitly through the confidence-width term: an action tried twice gets a large bonus and an action tried ten thousand times gets almost none, so the bonus is literally an uncertainty estimate.

And it explains why optimistic initialisation works at all. A value function initialised above every achievable value makes every unvisited state attractive, so exploration is driven by the value function itself rather than bolted on beside it — which means the exploration is *directed* toward what has not been seen, rather than being uniform noise. **That distinction is the whole content of §3.**

3 Deep exploration

A CORRIDOR of length n . Reward only at the far end.
Actions: left, right.

WITH EPSILON-GREEDY from a zero-initialised Q :
the agent performs a RANDOM WALK until it stumbles onto
the reward. The expected time is $O(2^n)$ -- exponential
in the corridor length.

THE PROBLEM IS NOT INSUFFICIENT RANDOMNESS. It is that
REACHING THE REWARD REQUIRES A LONG COHERENT SEQUENCE of
the same choice, and INDEPENDENT PER-STEP NOISE DESTROYS
COHERENCE BY CONSTRUCTION. More noise makes it worse.

THIS IS CALLED DEEP EXPLORATION, and it is the open problem
of the field.

PARTIAL ANSWERS

- bootstrapped DQN / randomised value functions
 - sample a Q -function from an approximate posterior at
the START OF EACH EPISODE and follow it greedily. The
randomness is now PER-EPISODE, so the behaviour is
coherent within an episode and different between
episodes. This is Thompson sampling, lifted to MDPs.
- parameter-space noise
 - perturb the policy's WEIGHTS rather than its actions.
Same reasoning: one perturbation produces one
coherent behaviour.
- count-based or curiosity bonuses
 - reward visiting rarely-seen states (section 4), so
exploration becomes directed rather than random.
- options / temporally extended actions
 - make the coherent sequence a SINGLE action, so no
coherence has to be maintained at all.

AND NONE OF THEM SOLVES MONTEZUMA'S REVENGE WITHOUT
DEMONSTRATIONS OR HAND-DESIGNED HELP, which is the honest
status of the problem.

Randomness must be coherent over time to explore deeply, and that single observation explains why
per-episode and parameter-space noise outperform per-step action noise despite being less random in total.
It also explains why §2's optimism matters so much: an optimistic value function produces coherent
directed behaviour (go to the unexplored place, consistently) rather than noise, which is the property the
corridor needs.

4 Intrinsic motivation

INTRINSIC REWARD, AND THE NOISY-TV PROBLEM

Add a bonus to the reward for visiting states that are novel — measured by an explicit visit count in small domains, a hash-based pseudo-count in large ones, or the prediction error of a learned dynamics model, on the reasoning that error is high where experience is scarce.

It works, and it is the best available answer for sparse-reward tasks — curiosity-driven agents reach rewards that ϵ -greedy provably never finds (§3), and the improvement on hard exploration games is large.

And the characteristic failure is the noisy TV. A source of genuine, irreducible randomness in the environment — a static-filled screen, a random number display, leaves moving in wind — is permanently 'novel' under a prediction-error measure, because the prediction error never falls however long you watch. So the agent sits in front of it forever, maximally curious and entirely useless.

Random network distillation addresses it neatly: predict the output of a *fixed, randomly initialised* network applied to the state. **That target is deterministic given the state**, so the prediction error falls with *familiarity* and not with *predictability* — and a noisy TV becomes familiar even though it stays unpredictable. **A clean fix for a specific real failure**, and a good example of diagnosing a failure precisely enough that the remedy follows.

- **Start with ϵ -greedy decayed to a small floor, or with the entropy bonus of Module 07 §3. It is usually enough**, and the elaborate methods are for tasks that demonstrably need them.
- **Diagnose before escalating. Plot state visitation. If the agent never reaches the rewarding region at all, exploration is the problem; if it reaches it regularly and still does not improve, exploration is not the problem** and you are looking at a learning or credit-assignment failure (Module 04). **These two cases call for completely different fixes and are routinely confused.**
- **Prefer fixing the reward over fixing the explorer. A denser reward signal is usually far cheaper than a better exploration algorithm** — with Module 10's warning firmly attached, because reward shaping is how agents learn to do the wrong thing efficiently.
- **Then try per-episode coherent noise (§3) before count-based bonuses**, which add a hyperparameter, a second network, and the failure mode above.
- **And log the exploration parameter every run. A collapsed policy entropy or a fully decayed ϵ explains a performance plateau immediately**, and is otherwise one of the hardest things to notice from a learning curve alone.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Sutton & Barto — chapter 2 (free PDF)	Bandits, §1's strategies, and the regret framing , with the experiments worth reproducing.
Lattimore & Szepesvári — Bandit Algorithms (free PDF)	Free in full, and the reference for §1 and §2 — the UCB and Thompson sampling analyses done properly.
Osband et al. — Deep Exploration via Bootstrapped DQN (free)	§3's corridor argument and the per-episode randomisation answer , from the paper that framed deep exploration clearly.
Burda et al. — Exploration by Random Network Distillation (free)	§4's fix for the noisy-TV problem , with the diagnosis that motivates it.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Implement a 10-armed bandit** and compare all five strategies from §1 over 2000 steps, averaged over 100 runs.
- 2 **Plot cumulative regret** and confirm ϵ -greedy's is linear and UCB's logarithmic.
- 3 **Implement optimistic initialisation** and report how long the optimism takes to wash out.
- 4 **Implement Thompson sampling** and compare against UCB.
- 5 **Build the corridor of §3** at $n = 5, 10, 20$ and measure ϵ -greedy's steps to first reward.
- 6 **Confirm the exponential scaling** and show that increasing ϵ makes it worse.
- 7 **Implement per-episode randomised value functions** and report the improvement on the same corridor.
- 8 **Add a count-based bonus** and compare.
- 9 **Build a noisy TV** into an environment and confirm a prediction-error bonus traps the agent.
- 10 **Implement random network distillation** and confirm it does not.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 What is a bandit, and why is it the right place to study exploration?
- 2 Define regret and state the best achievable rate.
- 3 Compare five exploration strategies.
- 4 Where else in this program does UCB's formula appear?
- 5 State the optimism principle and explain why it is self-correcting.

- 6 What is ϵ -greedy's real failure?
- 7 Why does per-step noise fail on the corridor, and what does the corridor cost?
- 8 Name four partial answers to deep exploration.
- 9 What is the noisy-TV problem and how does RND fix it?
- 10 What does a state-visitation plot distinguish?